

1. Web Lab (IT5PC_LR3)

1.1 Teaching Scheme

Course Code	Course Name	Contact Hours	Credits
IT5PC_LR3	Web Lab	02	01

1.2 Examination and Evaluation Scheme

Evaluation	ISE	PE	OrE	PrE	Total
Marks	25	0	0	25	50

1.3 Lab Objectives

Sr. No.	Objectives
1	To introduce open-source tools for web analytics and the semantic web and build basic HTML and CSS skills for creating styled web pages.
2	To enable students to use JavaScript for developing interactive web pages and apply TypeScript for building efficient and scalable web applications.
3	To introduce students to React for developing front-end applications and AngularJS for building single-page web applications.
4	To enable students to use Node.js for backend development and Flask for creating efficient web applications.
5	To enable students to work with MongoDB and Mongoose for database operations and schema validation, and to apply these skills in a mini project by designing a complete web application.

1.4 Lab Outcomes

The student will be able to	
1	Demonstrate foundational skills in HTML and CSS to create and style web pages.
2	Apply JavaScript to make web pages interactive and use TypeScript for developing structured and scalable web applications.
3	Develop front-end applications using React and build single-page web applications with AngularJS.
4	Implement backend functionality using Node.js and Flask to develop efficient web applications.
5	Perform database operations with MongoDB, design schemas using Mongoose, integrate them into web applications.

1.5 Suggested list of experiments

Sr. No.	List of Experiments
1	To study web analytics using open source tools like Matomo, Open Web Analytics, AWStats, Countly, Plausible.
2	To design a personal profile using HTML5 and CSS.
3	To validate registration form using HTML5, CSS and JavaScript control structures.
4	To Perform Inheritance example using TypeScript.
5	To set up a development environment and create a single-page React application using React Router.
6	To Create a simple HTML “Hello World” Project using Angular Framework and learn to apply ng-controller, ng-model and expressions in AngularJS.
7	To build a simple web server and perform file handling using Node.js.
8	To design Feedback Form using Flask.
9	To install MongoDB, perform CRUD operations, and use Mongoose to connect Node.js with MongoDB for schema design and validation.
10	Mini Project - To design a website.

1.6 Resources

1.6.1 Textbooks

1. Advanced Internet Technology, Dr. Deven Shah, DreamTech, 2014.
2. Rediscovering JavaScript: Master ES6, ES7, and ES8, Venkat Subramaniam, Pragmatic Bookshelf, 2018
3. Learning React Functional Web Development with React and Redux, 1st Ed., Alex Banks and Eve Porcello, O’Reilly, 2017
4. Flask Web Development: Developing Web Applications with Python, 1st Ed., Miguel Grinberg, O’Reilly, 2014
5. Getting MEAN with Mongo, Express, Angular, and Node, Simon Holmes, Clive Harber, 2nd Ed., Manning Publications, 2019

1.6.2 Reference Books

1. TypeScript Quickly, 1st Ed. Yakov Fain and Anton Moiseev, Manning Publications, 2020.
2. Pro TypeScript: Application - Scale Javascript Development, 2nd Ed., Steve Fenton, Apress, 2018
3. Node.js, MongoDB and Angular Web Development, 2nd Ed., Brad Dayley, Brendan Dayley, Caleb Dayley, Pearson Education, 2018
4. MEAN Web Development, Amos Q. Haviv, Packt Publishing, 2014
5. Web Development with Node and Express, 2nd Ed. Ethan Brown, O’Reilly, 2019

1.6.3 Web Resources

1. NPTEL course on Web Technology by Dr. Ashutosh Kumar Bhatt, Uttarakhand Open University, Haldwani
https://onlinecourses.swayam2.ac.in/e-learning/preview/nou25_cs09

2. Coursera course on Web Application Development with JavaScript and MongoDB by Dr Matthew Yee-King, University of London
<https://tinyurl.com/mpa2yfnt>
3. Udemy course on Node with React: Full Stack Web Development by Stephen Grider, Udemy course
<https://tinyurl.com/42fnk35h>